

MGM3615: CNS Tumour Methylation Classifier

Report Details	Specimen Information	Ordering Clinician
Sample ID / Order ID: 9999213 / 1774752	Specimen Site: Brain	Clinician: Dr. Roopesh kumar
Collection Date: 19 th May 2026	Specimen Received: FFPE Tissue Blocks [1]	Affiliation: MGM Healthcare-Chennai
Date Received: 19 th May 2026	Specimen Tested: 1270/26 A	Serviced By: NA
Report Date & Time: 16 th June 2026 12:11 PM	Tumor Content: 80 %	Report Status: Final

Clinical / Histopathological Summary: Early high-grade pediatric-type diffuse infiltrating glioma with astro-glial differentiation, IDH-, H3K27-wildtype, p53 positive, in a KCO *DBN1::BRAF* fusion-positive, MAPK-altered pediatric-type diffuse low-grade glioma. FISH test is positive for hemizygous monosomy deletion of *CDKN2A/B*.

METHYLATION CLASSIFICATION

A methylation-based classification tool computationally analyses the DNA methylation patterns observed in the subject’s tumor sample and classifies it into different categories of central nervous system (CNS) tumors based on the WHO guidelines for Classification of CNS tumors (5th ed., 2021). Methylation profiling should be considered in conjunction with other histopathologic, genetic and clinical data to reach a diagnosis.

Predictions	Bethesda v2 classifier score	Epignostix v12.8 score	Interpretation ^
Superfamily Low-Grade Glial/Glioneuronal/Neuroepithelial Tumours	0.992	0.989	✓
Class Supratentorial Midline Pilocytic Astrocytoma	0.991	0.981	✓

- ✓ Match (score > 0.9)
 - Suggestive (score 0.5 - 0.9)
 - ✗ No match (score < 0.5)
- ^ Kindly refer to the Appendix section for a detailed guide on the interpretation of the scores assigned for the predictions

CRITICAL MOLECULAR FINDINGS

IDH	TERT	Chr1p/19q co-deletion	EGFR	Chr7 gain/ Chr10 loss	CDKN2A	CDKN2B	WHO Grade
Wildtype	Not determined	Not determined	Not determined	Not determined	Not determined	Not determined	1

MGMT PROMOTER STATUS

Promoter	Status	Score *
MGMT	Unmethylated	0.047

*Score >0.358 = MGMT promoter is “Methylated”

CLASS SUMMARY

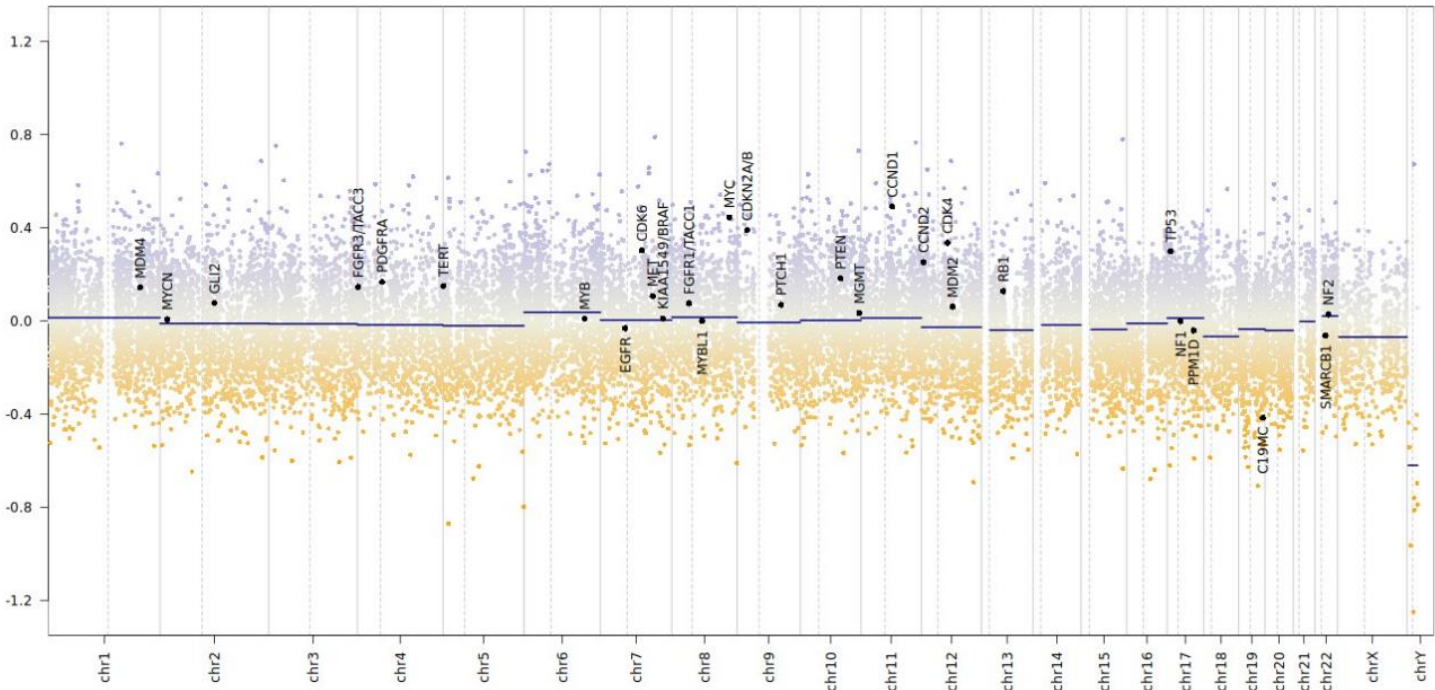
Superfamily/Tumor group: : Low-Grade Glial/Glioneuronal/Neuroepithelial Tumours

Summary: Low-grade glial and glioneuronal tumors encompass a highly heterogeneous group of slow-growing, primary central nervous system neoplasms. According to the WHO Classification of Tumours (5th ed., Vol. 6), these tumors span several distinct families, including circumscribed astrocytic gliomas, glioneuronal/neuronal tumors, and pediatric-type diffuse low-grade gliomas, and are uniformly classified as CNS WHO Grade 1 or 2. Unlike aggressive pediatric high-grade gliomas (pHGGs) driven by *H3* or *IDH* mutations, pLGGTs are primarily characterized by single, focal alterations that activate the MAPK/ERK signaling pathway. The most common drivers include the *KIAA1549::BRAF* gene fusion, *BRAF* V600E mutations, and various *FGFR1* or *MYB/MYBL1* alterations [PMID: 24664484, 35218102].

Class: Supratentorial Midline Pilocytic Astrocytoma

Summary: Pilocytic astrocytomas (PAs) are most often found in children and adolescents, decreasing in incidence with advancing age and account for approximately 15.6% of all brain tumors and 5.4% of all gliomas. They are mostly found in infratentorial structures such as the cerebellum and in midline cerebral structures such as the optic nerve, hypothalamus, and brain stem. Supratentorial midline PAs (sm-PAs) typically carry MAPK pathway alterations, most commonly *KIAA1549::BRAF* fusion, but also some others (for example *FGFR1* mutation is recurrently observed in this class and *NF1*-associated optic pathway gliomas typically belong to this category). In case of high quality data the *BRAF* duplication can be appreciated on chromosome 7q (present in >60% of the reference cases). In 81 subtotally resected (STR) PAs, no differences in PFS by location were observed between subjects with STR PAs at initial progression; however, those with sm-PAs exhibited shorter times to second progression. Children with sm-PA exhibited more neurologic deficits over time. sm-PAs exhibited higher rates of multiple progression, which correlates with increased neurologic morbidity [WHO Classification of tumours series, 5th ed.; vol. 6, PMID: 35036914].

COPY NUMBER PROFILE



(This CNV profile has been generated by Epignostix v12.8 classifier)

INTERPRETATION: In this case, the CNV plot shows a heterogeneous genomic profile and thus, definitive assessment of copy number alterations cannot be done.

The plot represents copy number changes across chromosomes 1-22 (and X/Y if automatic prediction was successful) (x-axis) in this subject with the log2 ratio of copy numbers on the y-axis. Green dots indicate copy number gains while red dots indicate copy number losses, and grey dots indicate no change in copy number. Segments are represented by blue lines, representing regions where the copy number is constant.

TEST DESCRIPTION

Medgenome’s CNS Tumour Methylation classifier is based on a methylation-microarray signature. This microarray-based central nervous system (CNS) tumor classification test identifies DNA-methylation-based patterns associated with different types of CNS tumors and detects various genomic alterations, including large copy number variations (CNVs) in CNS tumors.

TEST METHODOLOGY

CNS Tumor Methylation Classifier v12.8 [<https://app.epignostix.com>, PMID: [40492071](https://pubmed.ncbi.nlm.nih.gov/40492071/)] is trained based on 7,495 methylation profiles, comprising 184 sub-classes. Utilizing a Random Forest–based methodology, the classifier is rigorously validated through five-fold nested cross-validation, achieving a 95% subclass-level accuracy and a Brier Score of 0.028, indicative of well-calibrated probability estimates.

Bethesda V2 classifier (<https://methylscape.ccr.cancer.gov/analysis>) is based on a stratified ensemble classification approach which includes a two level classification, Superfamily, and Class. Each level of classification has multiple classifiers (n=10) based on functional or genomic regions related to methylation annotation: transcription start site (TSS), gene-body, shore and island etc. The mean calibrated score, calculated at Superfamily and class levels, is the most important it displays the average performance of all classifiers.

The large copy number prediction is performed using conumee2 program (Daenekas et al., 2024). *MGMT* promoter methylation probability was calculated by mgmtstp27 R package.

DISCLAIMER

- Details of genomic variations, including HGVS annotations, c. and p., Variant Allele Frequency (VAF), gene fusion breakpoints etc. are outside the scope of this methylation pattern based classification tool.
- Methylation profiling-based classification is intended for research use only; it has not been fully clinically validated and, therefore, should not be used for diagnostic purposes. Decisions regarding the treatment action plan should not be made based on these results.
- A low score, borderline score, or discrepant result may reflect technical limitations of the classifier, uniqueness of the sample, low tumor cell fraction, tumor heterogeneity, or tumor entities not well represented or not yet defined in the reference classifier cohort.

APPENDIX: INTERPRETATION OF CLASSIFIER SCORES

Both the levels of classifiers, Superfamily and Class, have multiple classifiers (n=10) based on functional/genomic regions related to methylation annotation. The mean calibrated calculated at Superfamily and class levels display the average performance of all the classifiers (n=10). **Below is an interpretation guide for the scores in the classifier. The interpretations “match”, “suggestive” and “no match” are indicative of the level of confidence with which the prediction has been made by the classifier tool.**

- Match = the tool has predicted with high confidence
- Suggestive = the tool has predicted with medium confidence
- No match = the tool has predicted with low confidence

Condition	Superfamily score	Class score	Interpretation
Condition 1	> 0.9	> 0.9	Superfamily - match / Class - match
Condition 2	> 0.9	0.5- 0.9	Superfamily - match / Class - suggestive
Condition 3	0.5- 0.9	0.5- 1	Superfamily - suggestive / Class - suggestive
Condition 4	0.5- 0.9	< 0.5	Superfamily - suggestive / Class - no match
Condition 5	< 0.5	any	Superfamily - no match / Class - no match

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END OF REPORT